

Multi-Agent Risks from Advanced AI

A Brief Overview

The Cooperative AI Foundation has released a research report entitled *Multi-Agent Risks from Advanced AI*. It explores the critical, underappreciated risks that arise specifically from AI agents interacting with each other and with people.

This document seeks to summarise the key findings and recommendations from the report for a non-technical audience.



Introduction

Most people's experience with AI so far has involved passive, static services such as chatbots and image generation tools. More recently, 'AI agents' have started to emerge. We might hear of them in the form of intelligent personal assistants, but such agents are already responsible for more high-stakes and consequential tasks such as [trading million-dollar assets](#) or [recommending actions to commanders in battle](#).



AI agents are AI systems that possess 'agentic capabilities'. The more they can operate in complex environments, pursue goals without being instructed, or act with little supervision, the more 'agentic' they are.

As AI developers pursue agentic capabilities in their models, the market incentives tied to automation gains will accelerate adoption across both industrial and public sectors. In order to fulfil their roles, these advanced agents will need to communicate and interact with each other and with people, giving rise to new multi-agent systems of unprecedented complexity.



Multi-agent systems consist of multiple decision-making agents which interact in a shared environment to achieve their goals, which may be consistent or in conflict with one another.

Why Should We Care About This Now?

- 1 **AI agents are already being deployed.** In the near future, we expect them to become more sophisticated and widely adopted. Crucially, they'll interact with each other, and the risks emerging from these complex interactions remain poorly understood. We urgently need to grasp the full spectrum of risks that may arise specifically from multi-agent behaviour.
- 2 **Our oversight and regulations are not keeping pace.** With advanced multi-agent systems on the horizon, our current frameworks for governance and oversight are falling behind. We lack both the widespread understanding to inform effective policy and the essential infrastructure needed to ensure safety.
- 3 **Foundational AI models aren't designed with multi-agent risks in mind.** We need to ensure they are built to cooperate safely and effectively within a world full of other AI agents.

Key Report Findings

- We've identified seven key mechanisms, or 'risk factors', that emerge **specifically from interactions between multiple AI agents**. These risk factors can lead to three high-level 'failure modes' in multi-agent systems: miscoordination, conflict, and collusion.
- These risk factors can – and do – emerge **even if single agents within the system are deemed safe**. We therefore need to develop specific methodologies and safety measures that account for the dynamic, interconnected nature of multi-agent AI systems.
- Tackling multi-agent risks will **require technical solutions combined with more societal approaches**, such as improved AI governance.
- The report outlines **concrete strategies for mitigating these risks**, from improved design principles for cooperative AI, to new regulatory approaches and promising research avenues.

How Multi-Agent Systems Can Fail

Multi-agent systems can fail in various ways, depending on the objectives of the agents and the intended behaviour of the system. The report identifies three distinct failure modes that emerge in multi-agent settings and can lead to significant risks.



Miscoordination

Miscoordination occurs when AI agents, despite having mutual and clear objectives, cannot align their behaviours to achieve these objectives. Agents might approach problem solving differently, or withhold sharing crucial information for the sake of faster decision making.

● EXAMPLE

Even if AI-powered driverless vehicles have perfectly valid strategies to keep passengers safe, they may fail to coordinate if their strategies are incompatible. The report provides a demonstration of how simple vision-language models implicitly trained on different driving conventions (such as yielding right versus left for emergency vehicles) can lead to catastrophic miscoordination when on the same road.



Conflict

In most real-world strategic interactions, AI agents will have objectives that are neither identical nor completely opposed, but mixed. Conflicts can arise and escalate due to the diverging interests of each agent in these settings.

● EXAMPLE

AI systems serving as advisors or negotiators in a military conflict can lead to rapid unintended escalation. When [Rivera et al. \(2024\)](#) tested AI agents controlling eight distinct nation-states in a military wargaming exercise, even neutral starting conditions did not prevent the rapid emergence of arms race dynamics and aggressive strategies. Strikingly, all five off-the-shelf LLMs studied showed forms of escalation, even when peaceful alternatives were available.



Collusion

AI collusion refers to unwanted cooperation between AI systems at the expense of other parties. This failure mode is particularly concerning because, unlike miscoordination problems that might naturally decrease with greater AI sophistication, collusion may become more severe as capabilities improve. Additionally, many promising approaches to ensuring the safety of advanced AI, such as adversarial training or oversight schemes, could be undermined by collusion between AI systems.

● EXAMPLE

AI systems controlling financial markets can collude at the expense of customers. [Assad et al. \(2020\)](#) showed how adaptive price-setting algorithms used at German gas stations in 2017 had a significant effect on fuel prices. They led to a 28% increase in profit margins in duopolistic markets and a 9% increase in non-monopoly markets. These results suggest they adapted their pricing strategies to collude, driving retail prices higher at the expense of consumers.

What Risk Factors Can Lead To or Make Failures Worse

To prevent these failure modes, it is necessary to consider the mechanisms via which they can arise or worsen. The report identifies seven key mechanisms or 'risk factors'. It's important to note that:

- 1 These factors are largely independent of the agents' precise goals and incentives.
- 2 These factors interact with and amplify each other, creating complex challenges that cannot be addressed by focusing solely on individual AI systems.

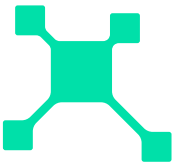


Information Asymmetries

When interacting agents possess different levels of information bearing on a joint action, they may face strategic incentives to withhold that information from others, or even mislead them in damaging ways.

● EXAMPLE

AI agents can learn to manipulate information about a financial market at the expense of other agents. [Shearer et al. \(2023\)](#) showed how an agent trained to maximise profit learned tactics to manipulate the financial benchmark, withholding its true state from other agents and making them worse off when trading.



Network Effects

As AI capabilities integrate into a wide range of existing networks, new risks emerge from the intricate relationships between individual components and the overall system.

● EXAMPLE

A single agent's manipulated knowledge can transfer across a network. [Ju et al. \(2024\)](#) demonstrated how an attacker could successfully modify an agent's internal parameters to treat false information as knowledgeable and persuasively spread this information across a network of benign agents. This corrupted information persisted in future LLM interactions through information retrieval.

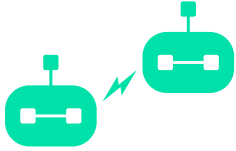


Selection Pressures

The way AI agents adapt and change over time due to external pressures (i.e., developer choices, co-adaptation between agents) and internal pressures (i.e., model training) can lead to concerning outcomes, such as competitive settings selecting for conflict-prone dispositions.

● EXAMPLE

LLMs accelerate resource depletion under pressures from the environment. [Piatti et al. \(2024\)](#) evaluated 15 different LLMs across three resource management scenarios. Even the most advanced LLMs achieved only a 54% survival rate, meaning that in nearly half of all cases, agents depleted their shared resources to the point of collapse.



Destabilising Dynamics

AI agents can adapt their strategies in response to other adaptive agents, at times resulting in damaging effects. We might observe this in the form of feedback loops or chaotic dynamics emerging as the number of learning agents increases.

● EXAMPLE

Feedback loops between agents can cause trillion-dollar losses. The 2010 Flash Crash, where algorithmic trading agents entered an unexpected feedback loop leading to a trillion-dollar market loss in minutes, illustrates how multi-agent systems can produce rapid, destabilising effects.

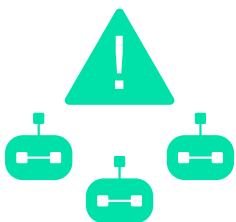


Commitment and Trust

Without trust or commitment ability, agents may fail to reach mutually beneficial agreements. But having commitment mechanisms that enable cooperation can also lead to new risks.

● EXAMPLE

Credible commitments can become dangerously dual-use. During the Cold War, the Soviet Union developed the automated Perimeter system – often called ‘Dead Hand’ – to guarantee a nuclear launch if its leadership were incapacitated, thus ensuring a credible commitment of retaliation. While this mechanism was intended as a deterrent, its automatic and largely irrevocable nature exemplifies how automated credible commitments can become dangerously dual-use: once triggered, there would be little chance to override or de-escalate.



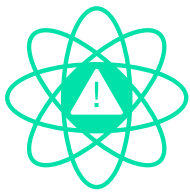
Emergent Agency

When AI systems are combined, novel forms of agency can emerge. For example, the collective might develop complex capabilities that go beyond that of each individual system, or pursue unintended outcomes as a group that no individual system was designed to seek.

● EXAMPLE

AI agents deemed safe can perform harmful tasks by sharing seemingly innocuous sub-tasks between them. [Jones et al. \(2024\)](#) found that agents that had been evaluated in isolation and assessed to be safe could

carefully delegate sub-tasks between them in order to successfully subvert their safeguards. For example, when attempting to generate vulnerable code, individual models in this experiment succeeded less than 3% of the time, while their combined approach succeeded 43% of the time.



Multi-Agent Security

The emergence of large numbers of advanced AI agents might – via their very multiplicity and decentralisation – lead to attack methods that would not be available to a single agent. Additionally, the complexity, interconnectedness, and range of such multi-agent systems may introduce new attack surfaces.

● EXAMPLE

An entire network of agents can be compromised by a single adversarial image. [Gu et al. \(2024\)](#) showed how a single adversarial image in a network of up to one million multimodal LLM agents can trigger ‘infectious’ jailbreak instructions that spread through routine agent-to-agent interactions, compromising the entire network.

What Can We Do About This Now?

Below we indicate some of the measures we can take to address multi-agent risks. While we provide some examples of how the Cooperative AI Foundation is addressing these, this is not an exhaustive representation of the efforts being put forward. We also believe much more action must be taken to prevent detrimental outcomes.

Support Research on Multi-Agent Risks

Significant research opportunities exist in the design, evaluation, and governance of groups of advanced AI agents. Alongside other institutions, the Cooperative AI Foundation supports this research via [grants](#) and [PhD Fellowships](#). However, more organisations ought to prioritise this important yet neglected area.

Perform AI Safety Evaluations That Factor in the Presence of Other Agents

This 'multi-agent testing' can help identify and mitigate potential risks. The Cooperative AI Foundation recently submitted an evaluation for agents' abilities to exploit and coerce one another to the UK AI Security Institute's [bounty programme](#), and will be developing further evaluations for dangerous capabilities and propensities later in 2025. We've also participated in consultations for the EU AI Act Code of Practice, and the NIST AI Risk Management Framework.

Build New Infrastructure for Interactions Between AI Agents

Just as internet protocols enabled the growth of the digital economy, building new infrastructure will be essential for realising the benefits of multi-agent AI while managing its risks. This could include using unique IDs to track AI agents, setting minimum interoperability standards, and developing tools for detecting anomalies and security breaches in large networks of agents.

Introduce Standardisation and Compliance Practices That Could Help Reduce Multi-Agent Risks

For example, standard documentation tools for single, static AI systems can be complemented with other forms of documentation that emphasise the connections between these systems and other key actors and resources, or the changing objectives and strategies of more agentic systems over time.

Reference

Hammond et al. (2025). *Multi-Agent Risks from Advanced AI*. Cooperative AI Foundation, Technical Report #1. <https://arxiv.org/pdf/2502.14143>

Who We Are

The Cooperative AI Foundation is a UK charity whose mission is to support and disseminate research on the cooperative intelligence of advanced AI systems for the benefit of all. To stay abreast of our initiatives you can:

Sign up to our mailing list ↗

Download the full Multi-Agent Risks from Advanced AI report ↓